Drone Surveillance for Fire Detection in Industrial Zones

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***Abstract*—In industrial environments, accidents can frequently occur due to manual inspections which causes a risk to workers safety. To address this problem, this paper introduces a drone-based fire detection system that make use of YOLOv8 and a Skydroid camera to improve safety and competence in monitoring. The system is mainly designed to identify fires in critical areas such as pipelines, boilers, and storage tanks easily and also reducing delays in emergency response. The drone captures real-time images, which are taken and sent it to the YOLOv8 algorithm. This algorithm checks each image we sent to it and provides a likelihood score to signify the chance a fire being present. This system not only improve the accuracy of fire detection and also reduces false alarms. Even it respond much faster and effectively covers high-risk areas in industrial zones. By using drones with AI powered software the system keeps workers safe and providing a responsible monitoring in complex industrial zones.**

Keywords—Industrial safety environments, Drone surveillance, You Only Look Once v8(YOLO), Hazard monitoring, Skydroid camera

I. INTRODUCTION

Industrial areas can be hazardous, making continuous safety monitoring is much difficult. These industrial environments are filled with heavy machinery, high temperatures, and flammable materials, which can considerably raise the risk of accidents, particularly fires which are caused by smoke.

In the past days, fire detection has depended only on manual checks and fixed sensors. Although, these methods only cover particular areas and might not detect fires quickly because of the smoke is spread over the area, potentially causing delays and serious issues.Over the years, industrial fires have tragically led to significant ruin including major or minor injuries, loss of life, environmental harm, and serious financial damage.

These drone based systems could operate in real-time and effectively reach hard-to-access areas, potentially making a big difference.To solve these problems, this paper presents an creative smart fire detection system that merge with a drone surveillance with image processing. By utilizing the YOLOv8 (You Only Look Once) deep learning algorithm, the system analyzes live video footage and even captured pictures also by the drones.

The YOLOv8 algorithm thoroughly checks the each frames captured by drone whether the fire is being present or not. This method make sure that quick and dependable monitoring in industrial settings, providing stress free and safety to the workers. Fig. 1 , represents that how accidents and worker deaths are caused by manual inspection in industrial settings.

To rectify this problem, the system mainly focuses on hazardous zones where more likely to catch fire.

Those zones like pipelines, boilers, storage tanks, and other. Drones are used to monitor the hazardous area where the fire can caught early and quickly alert security teams or automatic control systems. This system can help to reduce injuries.

In 2015, an unfortunate fire swept through a chemical plant in Shandong, China, that causes the loss of several lives of workers. Inquirers later revealed that the tragic incident was due to inadequate supervision and careless storage of hazardous chemicals.

In 2016, due to improper maintainence and delayed inspections, an oil pipeline in Saskatchewan, canada, a leakage of over 200,000 liters of crude oil spilling takes place.

In 2017 in Dhaka, Bangladesh in a factory due to using of old equipment a boiler exploded killing 13 workers. And it turned out to be a tragical incident.

In 2017, a boiler exploded at a garment factory in Dhaka, Bangladesh, killing 13 workers. It turned out the factory had been using old equipment and rarely carried out safety checks.

In 2018, due to failure of pressure valves and lack of inspections in oil refinery in Indonesia the refinery got blast causing heavy damage and several deaths.

In 2019, in Russia's Leningrad region due to lack of checking of chemical storage in a factory, a fire broke out in the factory leading to several deaths which is found by investigators.

In 2020, the tragic Visakhapatnam gas leak in India killed 13 people and injured many other peoples. The leak was caused by poor maintenance and skipped omitted inspections of the chemical tanks.

Then in 2021, a massive fire at a food processing plant in Bangladesh killing of over 50 workers. Locked exits and ignored safety checks turned what could have been a small fire into a major disaster.

In 2022, a massive fire broke out in a commercial building in Mundka, Delhi, claiming 27 lives. Investigations later revealed that the building lacked mandatory fire safety clearances and had not undergone recent safety inspections.

A year later, in 2023, an explosion at a fertilizer plant in Turkey was traced to severe corrosion in equipment that had gone unchecked for a long time, ultimately leading to the disaster.In 2024, a tragic explosion occurred at a pharmaceutical plant in Andhra Pradesh, India, resulting in several fatalities. The problem was caused by carelessness in checking and fixing the plant's electrical systems.

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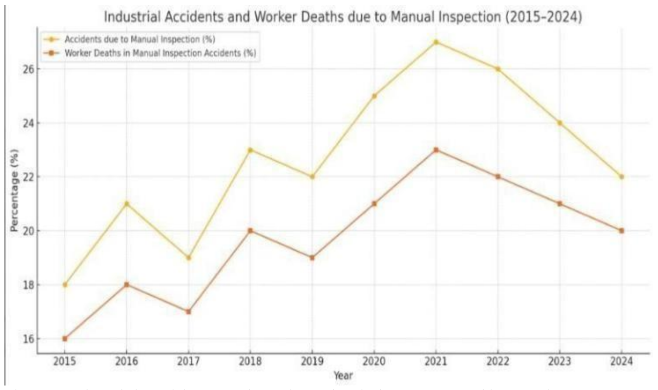

Fig. 1. Industrial accidents and workers death due to manual inspection.

II. LITERATURE SURVEY

J. J. Redmon and colleagues introduced the original YOLO algorithm, which completely changed real-time object detection by analyzing whole images in a single pass. In large industrial areas the early fire detection became difficult for YOLO although it is very fast, it is difficult for it to sense small fire and smoke [1].

Y. Wang and his colleagues later developed YOLOv8, which is an advanced model having increased accuracy and fire detection, and performs well on standard databases. But it should be still studied more for performing effectively in complex environments like industrial monitoring [2].

B. liu and his colleagues used YOLOv8 for outdoor fire detection in forests and grasslands and got succeeded for having high Accuracy, but the there are some setbacks regarding it's metallic body and the heat reflections [3].

A. Kumar and R. Singh mentioned clearly the value of drones in industrial safety and surveillance, but they concluded that most of the industries still depends on manual review in inspections. They suggested to combine AI-based fire detection techniques [4].

H. Rahman's report on Bangladesh factory fire explains the causes of delayed fire detection on people, and saying about giving the importance to early-warning systems for prevention of damage to people [5].

In the similar way The Hindu showed the Visakhapatnam gas leak using the drones saying the drones can play a crucial role in monitoring and covering the footage fast and accurately [6].

P. Sharma and K. Roy had reviewed on UAV's in industrial inspections and appreciates the drones abilities to reach the remote and dangerous areas, but also pointed that some, use AI systems for real-time applications in fire detection by reducing their power [7].

R. Mehta worked on traditional fire detection systems, suggests that combining of drones with AI can solve the problem of delayed responses and coverage gaps [8].

Zhang and M. Li explained about models like YOLOv8 are better at detecting small flames and smoke which are highly useful for drone monitoring, and he calimed that these are anchor-free models [9].

G. Xie and colleagues explained that for detecting fires in industrial areas, especially in hard and remote areas the development of strong deep learning models are necessary. They also said the importance of AI and drones combination [10].

L. Zhou and his colleagues explained the advantages of drones in wide coverage areas and fast setup but there are also some setbacks to face regarding the battery life, data delays and they can't detect small fires from high altitudes [11].

S. Singh and his colleagues explained the difficulties regarding the identification of small fires from fals alarms caused by heat reflections [12].

A. Behnam and his colleagues explained that the models like YOLOv8 are important and necessary for reducing the emergency response times, which are lightweight and efficient [13].

K. Sharma and his colleagues found that for improved fire detection, and faster work the YOLO-family models in drones are necessary [14].

Finally, D. S. Thompson et al. demonstrated YOLO and drones working together in hazardous environments, showing promising results but noting challenges like varying altitudes, lighting conditions, and dynamic industrial settings [15].

III. PROPOSED METHOD

A. Procedure For Inspection

Fig. 2 , Block diagram of the inspection process. Our UAV inspection service is suitable for manual operation in large open industrial areas where there is the greatest fire risk; such as those with fuel tanks, flammable pipework, valves and other hazardous materials).

Real-time data flow at specified time intervals a pilot drives the aircraft through predetermined routes to make sure they cover safety zones. he drone takes pictures of its surroundings in the air, and if it sees either suspicious signs (smoke or a visible flame) or fire before colliding with an object, the operator can screenshot that frame for closer inspection later.

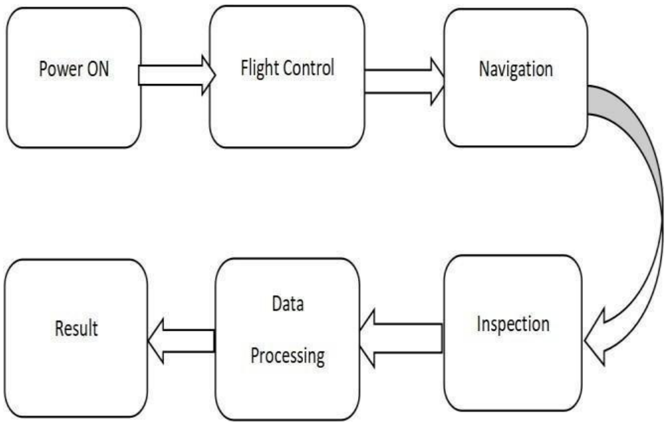

Fig. 2. Block Diagram of inspection Process.

Fig. 3 , illustrates the drone-based inspection process. The captured images are processed by the YOLO deep learning algorithm, which looks at visual features within each frame and generates a percentage confidence showing the extent there is a fire. This score tells how much fire is detected at that area as transparency to the operator.Unlike automatic fire-detection systems, which can sound alarms and summon emergency responders automatically, this doesn't do that all by itself. It is the operator's decision what to do with YOLOv8's results given the actual situation onsite." GPS tracking allows the drone's position to be tracked in real time for complete coverage of all inspection areas.Those captured images and also the detection results are stored for the future references, if the audits happens they use these images to boost and build upon your inspection system. This machine-assisted method is co-ordinates between the human and, making it a practical and flexible solution for hazardous industrial areas where accuracy and adaptability are essential.

Fig. 3. Drone Inspection Process.

B. YOLO(You Only Look Once)Algorithm

Our system combines both drone technology and fire detection. For a system using YOLOv8 deep learning model provide exact and adjustable monitoring method for outdoor applications. The process starts by manually operating drone at given time. These drones are operated at high risk areas such as fuel across storage tanks, pipes, pressure relief valves, and control junctions.Compared to automated monitoring systems this system provides the operator to modify the drone path and can make instant changes the plan according to the environmental conditions.During aerial inspection, the drone record high-resolution images from different angles over the target areas. These captured pictures are carefully examined, and showing possible signs of fire like smoke, heat distortion, or unusual activity are manually sent to the YOLOv8 model for analysis. YOLOv8, short for "You Only Look Once" version 8, is a modern object detection system that identifies objects accurately. Operators based on this score to make decisions high scores can send alerts to emergency teams or further visual checks, while lower scores can be also make corrections at that spot and also stored for reference or used to improve the model's training over time. Moreover, in this method YOLOv8 is used, by using this method the system allows to continuously learn and improve. Each and every image is processed by using this YOLOv8 method and the processed image, irrespective of whether it shows fire, can be saved, and used to refine the model's training for better accuracy in the future.This enables the development of a specialized dataset tailored to the specific visual characteristics and risk factors present in the user's own industrial environment. Over time, this process increases the model's reliability and sharpens its ability to distinguish between true fire events and visually similar non-hazardous conditions.

IV. MEASURING SETUP

A. Drone Fabrication

Fig. 4 , shows the drone arm with brushless motors, to configure the quadcopter's propulsion, begin by ensuring the frame is fully built, level, and unobstructed. Bolt each of the four brushless motors to the arm ends, applying a drop of thread-lock to prevent the screws from vibrating loose. Arrange the motors so that two spin clockwise and the other two spin counterclockwise, placing them diagonally opposite for balance. Secure each ESC near its motor or temporarily fasten it to the arm, keeping wiring neat to reduce voltage loss. Fit the propellers onto their shafts matching CW-marked props with clockwise motors and CCW-marked props with counterclockwise motors and hand-tighten them without over-torquing. the propulsion system is complete and ready for flight-controller integration and calibration. In this newly created file, highlight all of the contents and import your prepared text file.

Fig. 4. Drone arms with brushless motors.

Fig. 5, depicts the quadcopter base frame with the motors installed. After assembling the propulsion system, the next step is to set up the power supply and connect the flight controller (FC). Typically, a LiPo battery is used, either connected directly to the ESCs or routed through a Power Distribution Board (PDB), depending on the drone's design. The PDB acts as the central hub, distributing power from the battery to each ESC to make sure the motors receive the correct voltage. The PDB provides the power to other components such as the flight controller, receiver, and GPS if required.

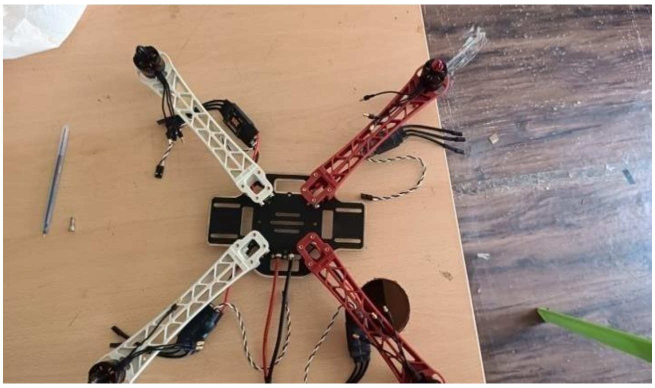

Fig. 5. Quad-copter Base Frame with Motor.

After placing the PDB, by using the ESCs signal wires the flight controller is connected to the ESCs which can be 3-wire PWM connections or digital links based on the setup. To reduce the motor induced vibrations the flight controller is placed at the centre of the frame. The receiver is then linked to the flight controller to receive remote control commands. Must check that all power connections are stable and that the wiring is neatly arranged to avoid loose or exposed connections that could cause shorts or interference and it is important also. After checking everything, the system is ready for calibration and testing, make sure proper power distribution and accurate communication between the flight controller and the ESCs.

Fig. 6 , shows the complete drone assembly, with the final steps involving the unification of the Skydroid system and preparation for the first flight. By securely connecting the camera to the drone frame after that make sure the camera visibility is clear. Connect the camera's power cables to the PDB, and link the video output to the transmitter. Keep the transmitter's antenna positioned away from other components to avoid interference. Then configure the Skydroid T10 receiver on the flight controller to control camera operations such as photo capture or video recording. At last, power on the drone and check the video feed on the ground station or FPV goggles, adjusting the camera angle as required.

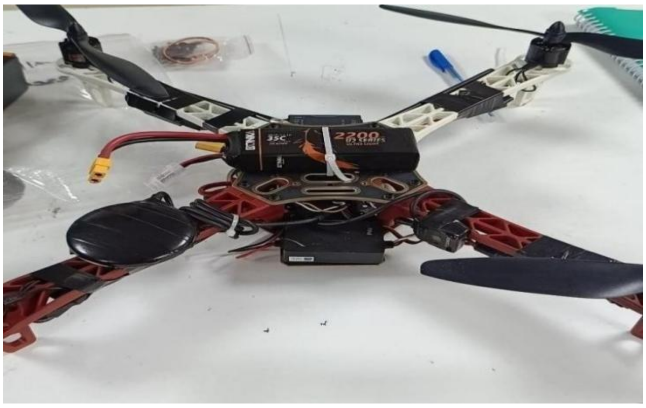

Fig. 6. Completed Drone Assembly.

After the video feed and control connections are confirmed to be steady, arm the motors and gradually increase the throttle in an open area to take off. During the initial flight test check the drone's stability, balance, and camera positioning. After reaching stable flight, perform basic maneuvers such as ascending, yawing, pitching, and rolling while monitoring the live video. When confident in the drone's handling, capture aerial footage. At last, reduce the throttle for a sleek landing, disarm the motors, and review the recorded footage to ensure that all systems operated correctly.

V. RESULTS

Fig. 7. Industrial Tank Inspection Using Surveillance Drone.

Fig. 7 , shows a tank inspection accomplished using a drone in an industrial environment. The drone-based fire detection system is designed to make sure that all critical areas are watched very closely for potential fire hazards. The drone is stabilized by its Flight Control Unit (FCU), is fitted with a medium-resolution skydroid camera that takes clear, high-quality images during flight. Its four propellers two spinning clockwise and two counterclockwise maintain stability and allow precise operation, enabling the drone to fly steadily over the inspection areas.

These areas typically include high-risk areas such as fuel storage tanks, pipelines, and valve systems, where the chances of fire risks are greatest.

Fig. 8. Fire caught on camera while inspection.

Fig. 8 , shows a fire detected by the drone's camera during an inspection process. The drone moves around the industrial area and captures the pictures by using skydroid camera. The operator carefully inspect these captured images to look for signs of fire, like smoke and visible flames. When potential fire indicators are dotted, the operator takes a screenshot and sends it to the YOLOv8 algorithm.

Fig. 9. Detection of fire using YOLO Algorithm.

The certain score reflects how likely it is that a fire is present in the image, giving image to the operator a clear, data-based assessment.Based on this fire percentage, the operator can then decide whether to calculate an overall detection percentage for the image, which represents the model's estimated probability of a fire being present. Fig. 9 , shows how the YOLO algorithm identifies fire in the secured frames. After the study, the system displays a particular percentage to the operator.

When this value is high enough to indicate a possible fire, the operator can either inspect the area more closely or alert the emergency response team for to take an immediate action.Images that show low confidence levels are also take an action to stop and also used to stored for later use, either for documentation or to enhance the model through further training. During the total detection process, additional information such as image time, GPS position, and detection results are recorded. These details help in maintaining exact tracking, reliable reporting, and continuous system improvement.

TABLE I. MANUAL VS PROPOSED METHOD COMPARISON

Aspect	Existing solution (manual)	Proposed Method (Drone+YOLOv8)
Operation	Workers physically inspect hazardous zones	Drone performs remote surveillance and captures real-time images
Accuracy	Relies on human judgment; prone to oversight and environmental limitations	More reliable with YOLOv8 providing probability-based fire detection
Response Time	Slower, as detection and reporting depend on manual effort	Quicker detection through automated image analysis in real time
Data Recording	Little or no systematic record-keeping	Automaticstorage of images and results for audits and model training
Human Safety	High risk due to direct presence in hazardous areas	Safer as monitoring is done remotely with minimal human exposure

VI. CONCLUSION

Detecting fire hazards quickly is essential in industrial areas, and using drones with the YOLOv8 model provides a modern and effective way to do it. Not only just by depending on traditional methods, this drone works on locating risky spots like boilers, tanks, and pipelines quickly. This drone is operated by a single person, while the identification of fires in the image is done by AI on other side. This Co-ordination between the human and the machine makes the work simpler, quick and reduces errors. With the help of GPS tracking, and detection large areas can be easily monitored by the system. The operator prevents the false alarms by reviewing the algorithm results before the starting of the action. For further improvement of model accuracy the saved images and data are used later. This semi-automated system provides worker safety and protects valuable property. As it collect more data, it continues to learn and react more effectively, helping industries stay safer and more reliable.

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